

Generation Interconnection Request System Impact Study Report

For Queue Project Q16-0719

Fairfield Works 100 kV
September 30, 2018

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q16-0719. The Interconnection Customer proposes a 168.3 MW solar photovoltaic facility interconnecting at the Fairfield Works 100 kV substation in the Illinois area. The request was received on December 27, 2016. The study evaluated system intact and single contingency conditions using the regional planning case, including 260 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 5 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$1,418,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

For the loss of Ravenna-Hylebos 500.0 kV, the Imperial-Chamberlin 230.0 kV loads to 113.7% of its 563 MVA rating. Project DFAX: 10.9%.

For the loss of Hylebos-Ravenna 500.0 kV, the Old Forge-Ravenna 230.0 kV loads to 103.7% of its 725 MVA rating. Project DFAX: 8.6%.

For the loss of Hilltop-Piney Creek Project 500.0 kV, the Sugar Hill Substation-Allegheny Energy Units 3 4 & 5 Substation 345.0 kV loads to 100.4% of its 1,709 MVA rating. Project DFAX: 9.9%.

Allegheny Energy Units 3 4 & 5 Substation 345-1 kV Autotransformer: 100.1% loading (1,558 MVA) following the outage of the Hilltop-Piney Creek Project 500.0 kV; DFAX = 9.6%.

For the loss of Piney Crk Project-Hilltop 500.0 kV line, the Allegheny Energy Units 3 4 & 5 500 kV/1 kV Transformer Bank loads to 100.1% of its 1,558 MVA rating. Project DFAX: 9.6%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Imperial-Chamberlin 230.0 kV	Rebuild line with higher-capacity conductor	\$15,245,000	2% (\$295,000)
Ravenna-Old Forge 230.0 kV line	Reconductor line	\$7,571,000	2% (\$146,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Allegheny Energy Units 3 4 & 5-Sugar Hill 345.0 kV	Reconductor line	\$4,858,000	2% (\$94,000)
Allegheny Energy Units 3 4 & 5 Substation 345-1 kV Autotransformer	Replace transformer with higher-rated unit	\$22,579,000	2% (\$439,000)
Allegheny Energy Units 3 4 & 5 500/1kV Transformer Bank	Replace transformer with higher-rated unit	\$22,822,000	2% (\$444,000)
	Total allocated to project		\$1,418,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.